

Netflix Content Analysis using SQL and Power BI

Chapter 1: Introduction

1.1 Introduction

Netflix is one of the world's leading streaming platforms, offering thousands of movies and television shows across different countries and genres. Due to the continuously growing content library, analyzing Netflix data manually is difficult. Data analytics tools help transform this large amount of information into meaningful insights.

This project analyzes the Netflix dataset using SQL and Microsoft Power BI. SQL is used for importing, cleaning, and querying the data, while Power BI is used to build an interactive dashboard that presents visual insights. The dashboard helps users understand content distribution, genre popularity, country-wise content, rating distribution, and content growth over time.

1.2 Objectives

- Analyze Netflix movies and TV shows.
 - Clean and prepare the dataset using SQL.
 - Create useful SQL queries for business analysis.
 - Build an interactive Power BI dashboard.
 - Generate business insights from the dashboard.
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Chapter 2: Project Overview

2.1 Dataset Description

The project uses the Netflix Titles Dataset obtained from Kaggle. The dataset contains information about Netflix content such as title, type, director, cast, country, release year, rating, duration, genres, and description.

Dataset Information

- Dataset Name: Netflix Titles Dataset
- Source: Kaggle
- Total Records: Approximately 8,800
- Total Columns: 12

Column Name	Description
show_id	Unique content ID
type	Movie or TV Show
title	Content title
director	Director name
cast	Cast members
country	Country of production
date_added	Date added to Netflix
release_year	Release year
rating	Audience rating
duration	Movie duration or TV seasons
listed_in	Genre
description	Content summary

2.2 Tools and Technologies

Tool	Purpose
MySQL	Data storage and cleaning
SQL	Data analysis
Power BI	Dashboard creation
DAX	KPI measures
Microsoft Excel	Dataset management

Chapter 3: Methodology

The project follows a structured data analytics workflow.

Step 1: Data Collection

The Netflix dataset was downloaded from Kaggle and imported into MySQL.

Step 2: Data Cleaning

The following preprocessing tasks were performed:

- Checked missing values
- Removed duplicate records
- Verified data types
- Cleaned inconsistent values
- Prepared the dataset for analysis

Step 3: SQL Analysis

Several SQL queries were written to analyze the dataset, including:

- Total Titles
- Total Movies
- Total TV Shows
- Top 10 Countries
- Top Genres
- Top Directors
- Content Rating Distribution
- Release Year Analysis

Step 4: Power BI Dashboard

The cleaned dataset was imported into Power BI. DAX measures were created, followed by KPI cards, interactive charts, and slicers to build a dynamic dashboard.

DAX Measures Used

- Total Titles
 - Total Movies
 - Total TV Shows
 - Movie %
 - TV Show %
 - Total Countries
 - Total Directors
 - Latest Release Year
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Chapter 4: Dashboard Analysis and Insights

The Power BI dashboard presents Netflix content through interactive visualizations.

Dashboard Components

KPI Cards

- Total Titles
- Total Movies
- Total TV Shows
- Total Countries
- Total Directors
- Latest Release Year

Charts

- Movies vs TV Shows Distribution (Donut Chart)
- Netflix Content Growth Over the Years (Line Chart)
- Top 10 Countries by Content (Bar Chart)
- Top 10 Genres (Bar Chart)
- Content Rating Distribution (Column Chart)
- Top Directors (Bar Chart)

Slicers

- Content Type
- Country
- Rating
- Release Year

Key Insights

- Movies account for the majority of Netflix content.
- The United States contributes the highest number of titles.
- Drama and International Movies are among the most common genres.

- TV-MA is one of the most frequently assigned content ratings.
 - Netflix's content library experienced significant growth after 2015.
 - The interactive dashboard allows users to filter content by country, rating, type, and release year for deeper analysis.
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Chapter 5: Conclusion and Future Scope

Conclusion

The Netflix Content Analysis project demonstrates how SQL and Microsoft Power BI can be combined to transform raw data into meaningful business insights. SQL was used for data cleaning and analysis, while Power BI provided interactive visualizations that make data exploration simple and effective.

The dashboard successfully highlights important trends related to content type, genre distribution, country contribution, content ratings, and release year. The project also demonstrates practical skills in SQL querying, DAX calculations, dashboard design, and business intelligence reporting.

Future Scope

- Integrate IMDb ratings for richer analysis.
- Build predictive models using Machine Learning.
- Connect the dashboard to live streaming data.
- Develop advanced recommendation analytics.
- Publish the dashboard using the Power BI Service for online access.

References

1. Kaggle – Netflix Titles Dataset
 2. Microsoft Power BI Documentation
 3. MySQL Documentation
 4. Microsoft Learn – DAX Documentation
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